

Poster Project Outline: Gene Expression in Short Term Spheroid Culture of High-Grade Serious Ovarian Cancer (HGSOC) Patients

Introduction

Of all ovarian cancers, around 85-90% of them are epithelial ovarian cancer. (Kim et al. 2018) High-Grade Serous Ovarian Cancer (HGSOC) is the most common subtype of the more aggressive type II epithelial ovarian cancer, accounting for 70% of all epithelial ovarian cancer. (Kim et al. 2018) Genetically, HGSOC can be characterized by its high genomic instability, especially in tumour suppressor TP53 as more than 96% of HGSOC samples collected contained some form of somatic TP53 mutation. (Lisio et al. 2019) Unfortunately, the disease is often diagnosed at advanced stages where cancer cells have metastasized, since in early stages, symptoms can be vague and not easily detectable. (Kim et al. 2018, Kordowitzki et al. 2025) Due to its later diagnosis, HGSOC is responsible for 73.9% of epithelial ovarian cancer deaths. (Kim et al. 2018) Effective treatment is necessary to reduce the mortality of HGSOC.

To increase treatment efficiency, reduce side effects, and improve patient survival, personalized medicine is often used for cancer patients. Patient-derived organoids (PDOs) can be used to develop genomic models to better understand the patient's disease or to test drug responses *ex vivo* to identify the most effective treatment for the patient. (Wensink et al. 2021, Vias et al. 2025)

Knowledge Gaps

The formation of multicellular spheroids is a massive complicating factor in treating complex cancers, conferring resistance to conventional chemotherapies and promoting recurrence. Current understanding of the development and behaviour of spheroids remains understudied. Spheroids also have the potential to more accurately model *in vivo* tumours to identify more effective, individualized treatments for aggressive and late-stage cancers (Porter et al. 2020). Our goal is to better understand the expression patterns of these short-term spheroid cultures to address some of these shortcomings.

Description of Data

The dataset consists of eight individual files of culture transcriptomics data collected from four patients at Day 0 and Day 6. Each file includes the Ensembl gene ID, the common name of the gene, and the read count of the gene. Since not much metadata is available on the dataset, we assume that culturing and treatment across samples were identical.

There are 5 variables present in the original dataset: patient, culture day, the Ensembl Gene Record and Common Gene Name (redundant variables, for the data analysis, only Ensembl Gene Record is used), and Read Count. All variables are independent and categorical except for read count, which is a dependent continuous variable.

Original Count data is heavily left-skewed with the majority of gene counts between . Performing a log fold correction to the read count shows that the distribution of log counts per gene has a bimodal distribution consistent among all files.

For more information regarding the data structure, distribution, and data cleaning/wrangling, please view PosterProject.py

All visualization was performed using R, .rmd file in 'Poster Project– Draft Visualizations'

Please view the following link:

<https://github.com/BestAbrar/BINF-5503-Group-5-Poster-Project>

Research Questions

For this project, the proposed research questions are:

1. Is there a significant change in gene expression (DESeq, padj value < 0.05) between days 0 and 6 of spheroid culturing?
2. If significant gene expression changes are observed (padj value < 0.05), are these changes significantly different between patients?
3. Do the most differentially expressed genes involve related pathways?

Hypotheses

For our research questions, we have developed the following hypotheses:

- 1) Null hypothesis - there will be no significant change in expression between days 0 and 6 of culture.
Alternative hypothesis - a significant change in gene expression will be observed after 6 days of spheroid culturing.
- 2) Null hypothesis - there will be no significant differences in expression between patient samples.
Alternative hypothesis - gene expression patterns will vary between patient samples.
- 3) Null hypothesis - differentially expressed genes will share no relation in function or cellular pathway.
Alternative hypothesis - differentially expressed genes will originate from connected pathways.

Predictions

Given that this dataset includes a massive number of quantified genes, there are several trends expected in the data. Genes involved in cellular adhesion will likely see increases in expression across all patients, as this family of proteins has been shown to influence the formation and structure of cancer spheroids (Micek et al. 2023; Molika et al. 2004). This could include genes such as FN1, CDH1, and ACTN1. Additionally, HGSOCs are characterized by increased expression of estrogen receptors and decreased expression of progesterone receptors. It therefore seems likely that the genes encoding these receptors (ESR1 / ESR2 and PGR respectively) would show the corresponding shifts in expression (Lisio et al. 2019). Another metric used to

differentiate HGSOCs from lower-grade lesions is a Ki-67 proliferation index, which we expect to be elevated in this dataset (Lisio et al. 2019). Cyclin E1 amplification is linked to the chemotherapy resistance of several prominent cancer types, most prominently HGSOCs, making it a likely candidate for differential expression in these patient-derived spheroids (Vias et al. 2023). A 2025 study on epithelial ovarian cancers found a unique behaviour in the typically overexpressed MYC. This study identified that while MYC is indeed upregulated in cancer lesions, upon cellular detachment into spheroids, the expression of MYC decreases as cells enter an acquired dormancy (Buensuceso et al. 2025). We therefore expect to see MYC expression decrease across all patients from day 0 to day 6. Finally, a 2023 study found a correlation between MYC and ZWINT expression, and given the role of ZWINT in proliferation and migration, we expect it to trend with MYC in spheroid development (Vias et al. 2023).

Proposed Methods

Differential gene expression analysis will be used to answer our research questions. This method identifies genes upregulation and downregulation after 6 days of incubation, allowing us to conduct further studies on genes with the most differential expression pattern after incubation. The first step of our analysis involves data cleaning and filtering. The files contain rows of “__no_feature”, “__ambiguous”, “__too_low_aQual”, “__not_aligned”, “__alignment_not_unique”, which are not representative of the expression data in this project, thus removed before analysis. Genes with 0 read count on both Day 0 and Day 6 across all patients will be filtered out, as well as the genes with consistently low to no expression. Further data cleaning might be required as our analysis proceeds.

In this study, we will be using DESeq 2 for differential gene expression analysis. We plan to examine several design formulas for this analysis, first examining ~ day to observe expression changes that occur between the two measured timepoints. We will then compare the results of that analysis with a subsequent test for ~ patient + day to determine whether there are patient-specific effects on changes in gene expression. If the results from those two analyses differ significantly, we will also observe ~ patient to determine which genes show significant differences between patients. Because this dataset examines a massive array of genes, we will need to account for false discovery rate (FDR) by observing adjusted p-values built into the DESeq2 package. For data cleaning, processing, and plotting, we will be using pandas, numpy, and ggplot. We plan to incorporate tables showing the most significant differentially expressed genes based on patient and day cultured, scatter plots plotting the principle component analysis to visualize the gene expression, patient and culture day, a volcano plot, and heatmaps of differentially expressed genes to effectively visualize findings. We will also use an MA plot to observe the average gene expression level against its average change in expression between Day 0 and Day 6.

There are some anticipated roadblocks in our study. There are only two time points (Day 0 and Day 6) with minimum information in-between, and with only four patients in this study, there is low statistical confidence that the sample population is representative. As

previously mentioned, the treatment information on the short-term spheroids culture is unclear, so assumptions have to be made, and it might affect our findings. Another possible challenge would be in determining whether expression patterns are unique to the spheroids or general expression patterns for HGSOC.

Batch effects in this dataset may include differences in RNA extraction methods, library preparation batches, sequencing runs or flow cells, RNA quality differences, sequencing depth variation, and technician or reagent variability. If time points were processed in separate batches, this would introduce a confounding effect where technical variation could be mistaken for biological changes due to culture conditions. This was investigated using both statistical methods (DESeq) and visually with the PCA plot. If there is a reasonable assumption that batch effects are present in this dataset, batch correction will be used.

Selected Figures

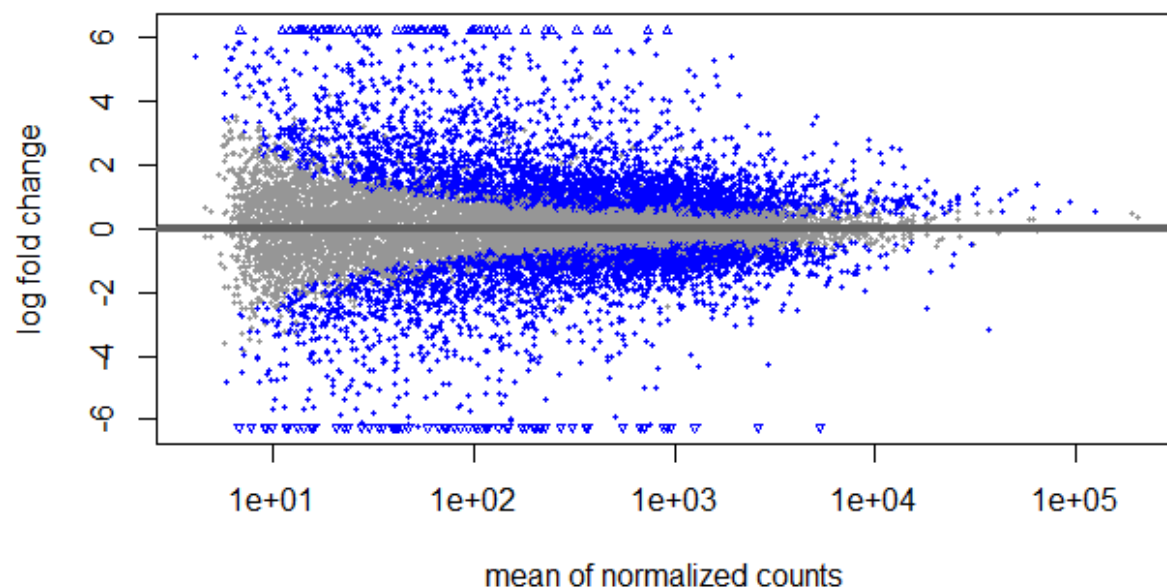


Figure 1) MA plot of differentially expressed genes in patient samples. Up- and down-regulated genes were shown in the top and bottom of the graph, respectively. Blue dots indicate the significant differentially expressed genes.

This MA plot helps to visualize the number of genes that have higher and lower average expression among all patient samples. It plots this against the log fold-change of these genes to provide an idea of whether high- or low-expression genes exhibit the largest change in expression level between day 0 and day 6 of spheroid culture. In this plot, there is a general trend where genes with lower base expression levels experience the highest changes in expression during the time course experiment. This means that the genes most impacted by the

culturing process are those that are not typically highly expressed in the tumours. While we cannot draw conclusions from this alone, it may indicate that there are niche subsets or families of genes being affected by the culturing process. For example, this could indicate cultural instability if these are stress response genes, or it could be adhesion-related genes being overexpressed as a response to the spheroids clustering.

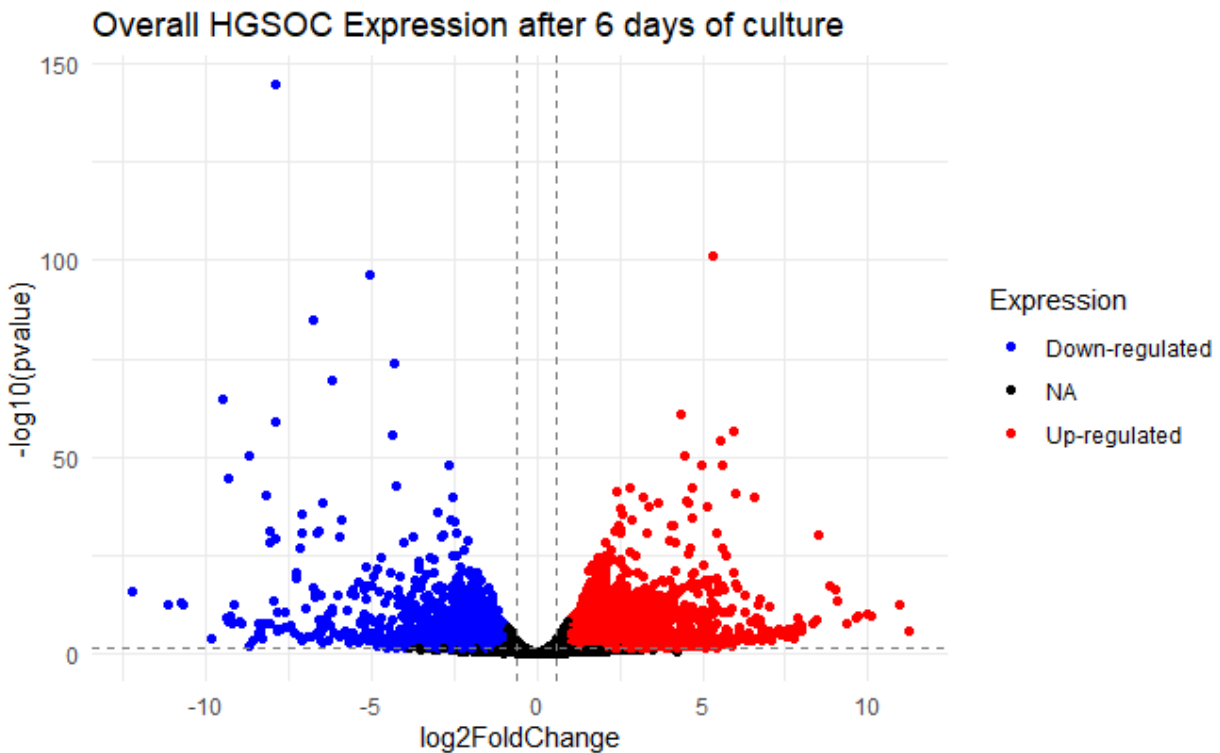


Figure 2) Volcano plot of differentially expressed genes. The down-regulated genes were shown on the left of the plot, indicated in blue dots. The up-regulated genes were shown on the right of the plot, indicated in red dots. Up/down regulation separated by ± 0.05 p-value respectively, and greater than $-\log_{10}(0.05)$

The volcano plot expands upon the previous MA plot by showing the relative significance of the highest differentially expressed genes. Here we see that many of the down-regulated genes show higher significance than up-regulated genes. While the general magnitudes between the up- and down-regulated genes are comparable, it appears that there is a more significant change with the down-regulated genes, with one gene in particular showing an approximately 7-fold decrease in expression with an extraordinarily low p-value. It will be worth looking at the genes that disperse from the main cluster (around the 50 mark on the y-axis) further to determine potential relationships.

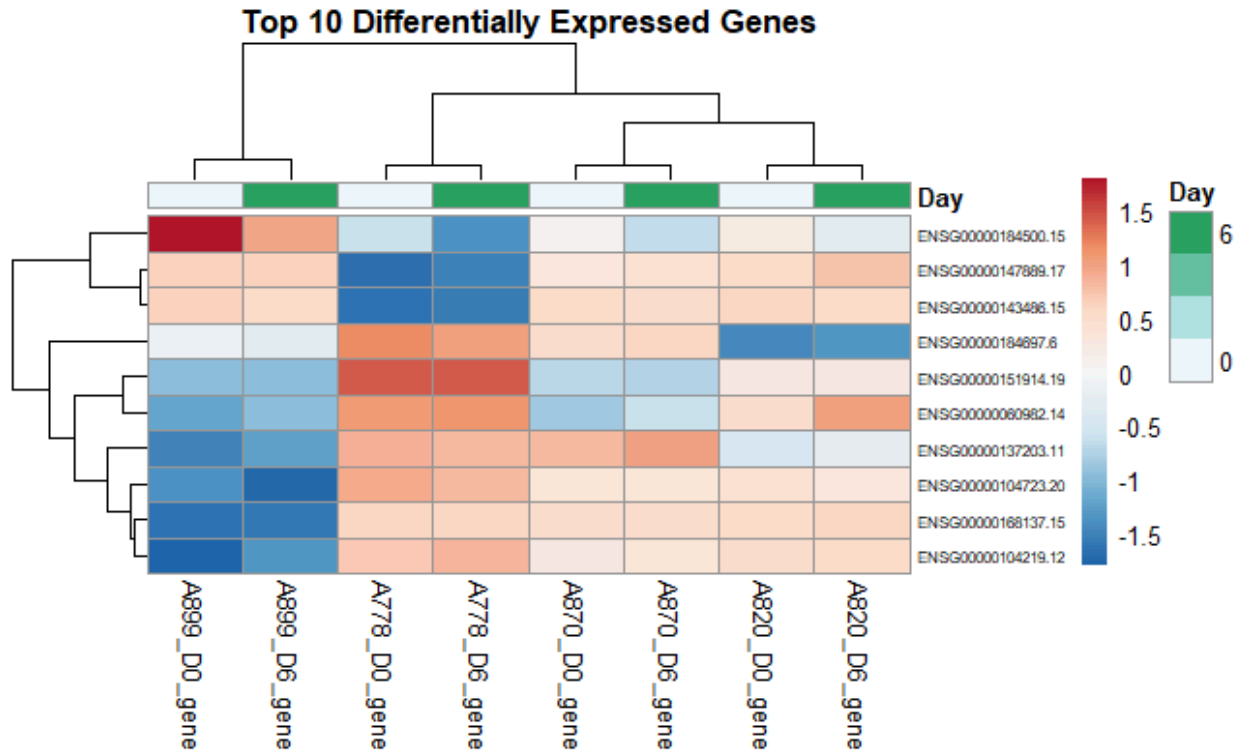


Figure 3) Clustered heatmap of the top 10 differentially expressed genes. Rows represent individual genes and columns represent the sample of origin, at Day 0 (white) and Day 6 (green). Red indicates up-regulation of the gene, while blue indicates down-regulation of the gene.

The clustered heatmap provides a visualization of the top 10 differentially expressed genes across patients at Day 0 and Day 6. Overall, gene expression patterns are more similar within the same patient across the two time points. Moreover, clear differences in gene expression patterns can be observed between different patients. For example, patient A870 and A820 are more closely clustered as the expression patterns are more similar. Samples from A899 showed 3 up-regulated genes and 6 down-regulated genes, while more up-regulation was observed than down-regulation in other patients.

ENSEMBL	SYMBOL	GENENAME	DEG
ENSG00000168137	SETD5	SET domain containing 5	Down-regulated
ENSG00000147889	CDKN2A	cyclin dependent kinase inhibitor 2A	Up-regulated
ENSG00000151914	DST	dystonin	Down-regulated
ENSG00000060982	BCAT1	branched chain amino acid transaminase 1	Down-regulated
ENSG00000184697	CLDN6	claudin 6	Down-regulated
ENSG00000137203	TFAP2A	transcription factor AP-2 alpha	Down-regulated
ENSG00000104723	TUSC3	tumor suppressor candidate 3	Down-regulated
ENSG00000184500	PROS1	protein S	Up-regulated
ENSG00000104219	ZDHHC2	zinc finger DHHC-type palmitoyltransferase 2	Down-regulated
ENSG00000143486	EIF2D	eukaryotic translation initiation factor 2D	Up-regulated

Table 1) Ensembl IDs, symbols, and names of top 10 differentially expressed genes. The table also indicates whether the gene is up-regulated or down-regulated (DEG).

Among the 10 most significantly differentially expressed genes, CDKN2A, PROS1, and EIF2D are up-regulated, while SETD5, DST, BCAT1, CLDN6, TFAP2A, TUSC3, and ZDHHC2 are down-regulated.

Gene	Description	Related Pathways	Associated Diseases	Interacting Proteins
PROS1	Produces protein S, a vitamin K-dependent plasma glycoprotein	Blood clotting	Thrombophilia	C4BPB, F5, F8
CDKN2A	Ubiquitously expressed protein that can block the cell cycle and activate apoptotic proteins	Apoptosis, Cell cycle regulation	Numerous cancers, Li-Fraumeni Syndrome	MDM2, CDK6, CDK4, TP53, NPM1
EIF2D	A translation initiation factor that delivers tRNAs to the P-site of the ribosome	Translation	No current strong associations are reported	RPS3, RPS11, RPS16, RPS4X, RACK1
CLDN6	As a member of the Claudin family of genes, it plays an important role in regulating tight junctions	Cellular adhesion	Some studies report links to breast cancer	TJP1

DST	Junction protein that links intermediate filaments to the actin cytoskeleton	Cellular adhesion	Neuropathy, Epidermolysis	COL17A1, MAPRE1, CALM1
BCAT1	Catalyzes transamination of branched-chain α -keto acids to branched-chain L-amino acids	Cellular metabolism, Peptide chain elongation	Hyperleucine-Isoleuc inemia, some studies report its overexpression in cancer (Li et al., 2022)	ARNT
TFAP2A	Transcription factor that promotes expression of a wide range of genes	Transcription, Cellular proliferation, Tissue development, Cellular adhesion	Branchiooculofacial Syndrome, various disorders associated with facial development	Not clear
TUSC3	Necessary component of the N-oligosaccharyl transferase complex	Translation, Protein N-Linked Glycosylation	Intellectual Developmental Disorder, is a candidate tumour suppressor gene	DAD1, DDOST, RPN2
SETD5	Histone H3 Methyltransferase	Histone Methylation	Intellectual Developmental Disorder	PRMT2, SKAP1
ZDHHC2	Palmitoyltransferase	Translation	Neuronal Ceroid Lipofuscinosis	Not clear

Table 2) Functional annotation of top 10 differentially expressed genes. Information including description, related pathways, associated diseases, and interacting proteins are summarized.

After determining the top 10 differentially expressed genes, further research was conducted to better understand the genes. Findings are summarized in Table 2, a brief description, the related pathways, associated diseases, and major interacting proteins are listed for each of the 10 genes. Here, we can use this information to begin making connections between the changes in expression and cultural health. CDKN2A is up-regulated in the patients, which is reasonable considering its role as a cell cycle regulator with numerous links to tumorigenesis. In addition, down-regulated genes CLDN6 and DST are both involved in tight junctions and cytoskeletal interactions, which may be relevant for spheroid clustering.

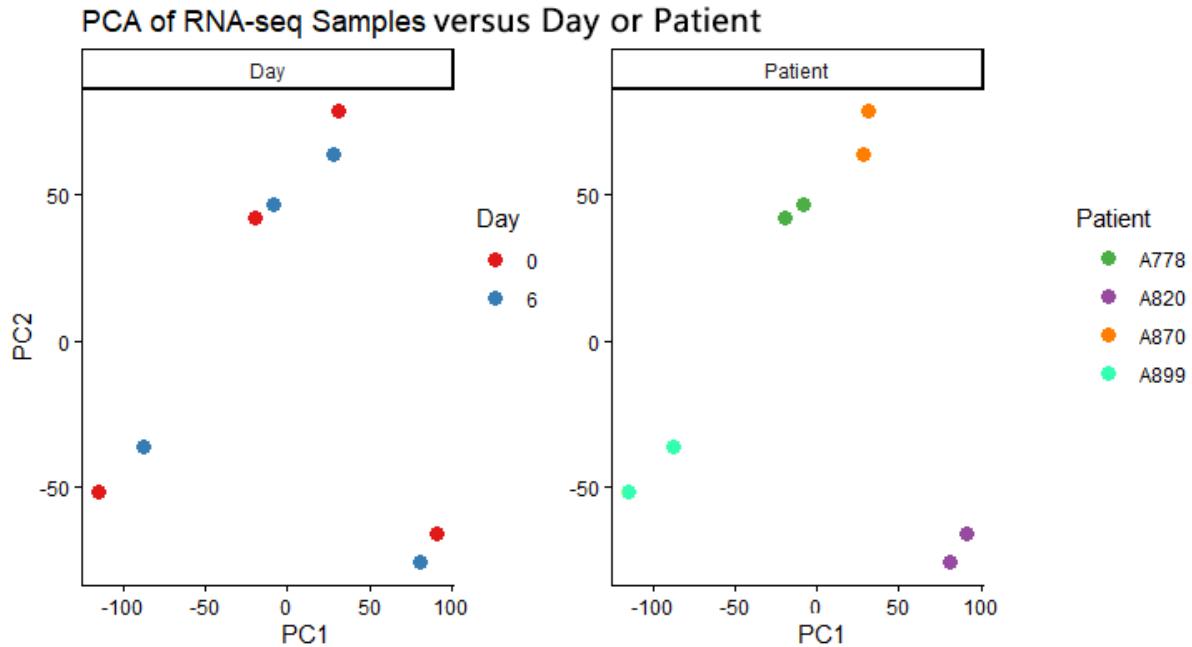


Figure 4) Scatterplot showing the PCA for each sequencing run based on the count/expression data of 17240 genes. Each plot (Day and Patient respectively) show identical points based on the first and second principal components, which explain 55% of the total variance observed (Figure 5).

Based on the PCA plot for all sequencing runs (Figure 4), cultures from patient A778 and A870 are more similar to each other than A820 or A899. Day 0 and 6 are also not significantly correlated, forming pairs rather than Day 0 grouping together with other Day 0 and Day 6 grouping together with other Day 6 points.

Principal component analysis revealed that samples clustered primarily by patient rather than by culture day, indicating that RNA expression is more influenced by the patient rather than by culture day. Within each patient cluster, Day 0 and Day 6 samples were separated, suggesting gene expression changes during spheroid culturing but this change isn't consistent between patients. This pattern is in agreement with the pattern seen in the top 10 differentially expressed genes seen in Figure 4.

Both Figure 3 and 4 show that there isn't a significant batch effect in this dataset, because although clustering is observed in the PCA, the pattern is consistent with patients rather than by day; therefore, all runs were treated the same. Each patient is observed to cluster separately, which is consistent with the nature of spheroid HGSOc samples (Kim et al. 2018, Porter et al. 2020).

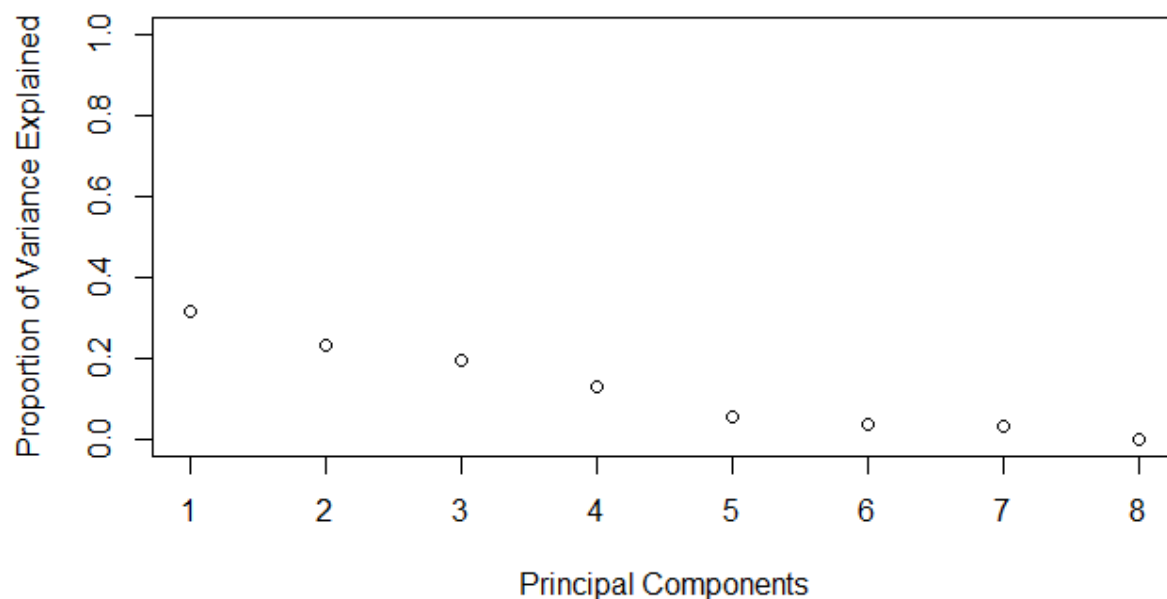


Figure 5) Visualization of the percent variance explained by each component in the principle component analysis performed in figure 4.

This plot shows the proportion of variance explained by each principal component. This helps determine which are the most relevant to the data and its exploration. We can see that over 50% of the total variance is explained by the first two principal components, while approximately 75% is explained by the first three. Using this we can determine that most of our focus in the analysis should be on the first three principal components to cover most of the complexity in the data.

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